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3rd year 1st Semester B.Sc. (Engg.) Final Examination-2022

Course Title: Data Mining
Course Code: ICT-3101
Time: 3 Hours
Marks: 70

N.B: i) Answer any five from the following questions
ii) The sequence should be maintained in answering each of the questions

1 a) List and explain the most significant actions that should be carried out during data preprocessing. 3

b) Give a good example to illustrate how binning methods works for data smoothing. 5

c) Suppose that a hospital tested the age and body fat for 14 randomly selected adults with the following results:

Age	23	23	27	27	39	41	47	49	50	52	54	58	56	57
%fat	9.5	17	26.5	17.8	31	8.5	34	32	33	42	28	27	61	60

Now calculate the mean, median, mode & SD of age & % fat 6

2 a) Write down the pseudocode for the frequent itemset generation part of the Apriori algorithm. 5

b) Illustrate the Apriori algorithm based on the given dataset:

Draw a plot based on their values.

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk

sup = 40 %
conf = 70 %

5

c) What is Simpons paradox? How do you identity Simpons Paradox? 4

3 a) Briefly described the basic architecture and the learning rule of a perceptron. 5

b) How does KNearest Neighbor (KNN) handle categorical features in the data-set? 4

c) How does KNN algorithm perform in comparison to other classification algorithm? Explain with example. 5

4 a) Prove or disprove the statement: if an itemset is frequent, then all its subsets must also be frequent. 4

b) Assume that there are only five items in the data set. Consider the set of 3-itemsets: {1, 2, 3}, {1, 2, 4}, {1, 2, 5}, {1, 3, 4}, {1, 3, 5}, {2, 3, 4}, {2, 3, 5}, {3, 4, 5}. Now solve the following- 5+

i. List all candidate 4-itemsets obtained by a candidate generation procedure using the $F_{F-1} X F_1$ merging strategy. 5

ii. List all candidate 4-itemsets obtained by the candidate generation procedure in Apriori.

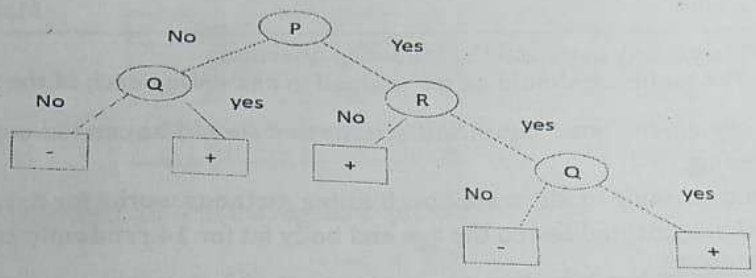
5 a) Few rules are given below:
R1 :(Gives Birth=no) ∧ (Aerial Creature=yes)→Birds
R2:(Gives Birth=no) ∧ (Aquatic Creature=yes)→Fishes
R3:(Gives Birth=no) ∧ (Aerial Creature=no)→Reptiles
R4:(Gives Birth=yes) ∧ (Body Temperature=warm-blooded)→Mammals
R5:(Aquatic Creature=semi) → Amphibians

Applying these rules, find out the type of creatures given in the following table:

Name	Body Temperature	Skin	Gives Birth	Aerial	Aquatic	Type
Grizzly bear	Warm-blooded	fur	yes	no	no	?
Hawk	Warm-blooded	feather	no	yes	no	?

4

b) A decisions tree is given below



5+
2

i. What are the rule sets for this tree?
ii. Can you simplify these rules derived in 4(b)(i)

c) What are the properties of a rule set? 3

6 a) State the bayes theorem with all the terms well explained. 2
b) What are the prior probability and posterior probability? 3
c) For a data set, x=110, S^2=2975, and S=54.54. If a test instance x is 120, find the candidate probability for a given class=NO. 4
d) let P(A=1) denotes the probability that Alam going to the campus while p(A=0) denotes the probability of him not going to the campus, with P(A=1)=40%. Also consider P(M=1|A) denote the conditional probability of Matin going to the campus conditioned on whether Alam is going to the campus. Thus, we have P(M=1|A=1)=80% and P(M=1|A=0)=30%. Find the probability that Alam is going to campus, given that Matin is going to the campus too. 5

7 a) Define (i) True positive rate (II) False positive rate 3
b) Consider the following confusion matrix

		Predicted class	
		+	-
Actual	+	100	100
Class	-	100	800

8

Calculate (i) F1 measure (ii) G Measure
c) How does the area under the ROC curve (AUC) determine whether the classification is perfect or perform random guessing? 3

8 a) Consider an ensemble of 25 binary classification, each of which has an error rate of €=0.25 Calculate the error rate of the ensemble classifier. 4
b) What are the steps in constructing an ensemble classifier? 3
c) What is the procedure of training of a random forest classifier? 3
d) How are rules generated or learned in a rule-based classifier? 4